

Beyond Solvable Models:

A Numerical Statistical-Mechanics Atlas of Modern Machine Learning

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Abstract

Modern machine learning contains recovery thresholds, specialization, trajectory changes, optimizer-dependent regimes, and collective alignment, but these phenomena are usually studied with incompatible objects and evidential standards. We present a numerical statistical-mechanics atlas that separates seven layers—domain, state object, deformation, disorder, scale, observable, and phenomenon—and records theory status, realism, and evidence on independent axes. A machine-readable PhaseCard and one experiment grammar connect a scientific question to an ensemble, finite-size path, intervention, estimator, and falsifier.

Transformers form the deepest dossier: trainable synthetic experiments measure attention, feed-forward, residual, objective, optimizer, scaling, and data coordinates with independent outer-seed uncertainty and separate intensive train, IID, and OOD risks. Frozen follow-ups test selected mechanisms without turning exploratory screens into confirmatory evidence. Diffusion, reinforcement learning, and multi-agent systems test whether the grammar remains useful with trajectory, feedback, and population observables. Predictive phase continuation is one atlas protocol: a calibrated surrogate must forecast an untouched size or deformation against simple baselines, with censored boundaries retained as outcomes. The contribution is a falsifiable map from questions to artifacts and negative results. It does not assert a shared universality class, nor that a smooth learning curve is a thermodynamic transition.

Reader’s guide and epistemic badges

This paper supports three reading paths. For the conceptual atlas, read Sections 2–4. For the statistical contract and predictive protocol, read Sections 5–6. For domain results, begin with the Transformer and diffusion dossier and then compare the RL, multi-agent, and predictive-continuation dossiers.

Every central claim can be read as three independent coordinates. Fidelity is **P phenomenological**, **S trainable synthetic**, or **N natural/pretrained**. Theory status is **T exact**, **A asymptotic**, **G phenomenological**, or **E empirical**. Evidence is **E0 descriptive** through **E6 externally replicated**. These badges are deliberately not collapsed into one rank: a large numerical study can remain synthetic, and an exact result can remain far from a realistic endpoint.

1 Introduction

Statistical mechanics offers machine learning more than analogies to phase transitions: it offers explicit ensembles, scale coordinates, observables, interventions, and falsifiable statements about collective behavior. Yet a phase diagram drawn after the fact can merely summarize a simulator, and the same word “entropy” can denote token uncertainty, macrostate multiplicity, or trajectory diversity. We therefore study one umbrella question:

How can numerical statistical mechanics describe, compare, and falsify collective phenomena across modern learning systems without confusing theory, realism, and evidence?

Our answer has three components. First, a seven-layer atlas states which state object, deformation, disorder ensemble, scale path, observable, and phenomenon are under study in each domain. Second, an experiment grammar maps these nouns to executable specifications and typed estimates. Third, domain dossiers state what was observed, what it may mean, and what remains unestablished. *Predictive phase continuation* is a particularly demanding protocol inside this grammar: calibrate at an anchor, predict an untouched deformation, diagnose breakdown, and test a frozen repair on independent evidence.

The domain coverage is deliberately asymmetric. Transformers receive the deepest mechanism study; diffusion, reinforcement learning (RL), and multi-agent systems test portability of the grammar. We do not propose one universality class for these systems. The atlas instead separates a genuine crossing from a censored threshold, an $O(1)$ risk from an extensive resource, temporal fluctuations from disorder fluctuations, and a five-seed screen from a finite-size phase claim.

The contributions are:

1. an orthogonal seven-layer taxonomy, observable dictionary, 13-outcome vocabulary, and PhaseCard v3 that preserve domain-specific physics;
2. a common execution and evidence grammar with typed estimates, nested disorder, explicit censoring, interventions, provenance, and invalidation;
3. a held-out predictive-continuation protocol with nontrivial baselines and prospective largest-size evaluation;
4. a Transformer flagship study that distinguishes broad exploratory coverage from frozen confirmation of module, optimizer, scaling, and data hypotheses; and
5. diffusion, RL, and multi-agent dossiers, plus literature, question, implementation, claim, and artifact ledgers that distinguish unknown, unimplemented, run, invalidated, and supported cells.

2 An orthogonal atlas of numerical statistical mechanics

The design space is a graph rather than a fully observed Cartesian tensor. A node is a scientific specification; an edge applies one registered deformation. Seven layers prevent axes of different abstraction from being conflated.

Domain. The domains are Transformer representations, diffusion trajectories, RL feedback dynamics, and multi-agent populations.

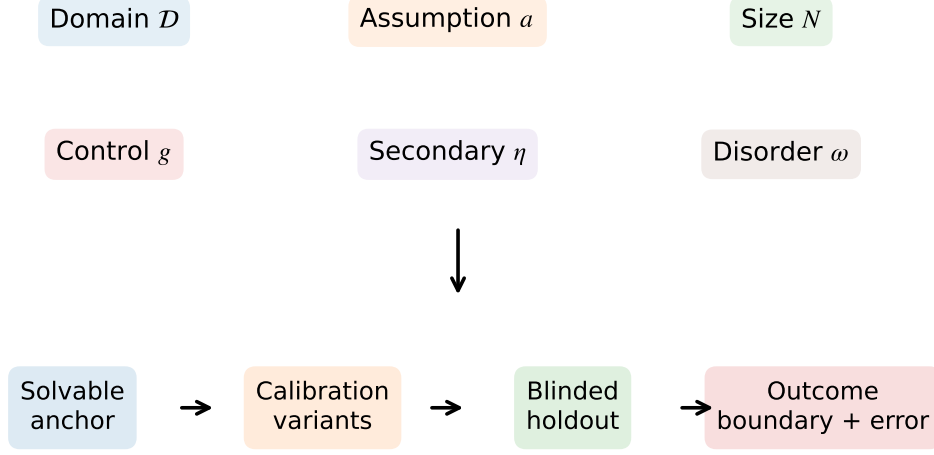
State object. The primary object is a parameter ensemble, representation, optimizer trajectory, generated sample, policy occupancy, or interacting population.

Deformation. Data/environment, architecture/interaction, objective/feedback, dynamics/optimizer, and lifecycle changes are explicit operators. Attention and MLP choices are subcoordinates of architecture, not peers of data or scale.

Disorder. Initialization, dataset draw, environment instance, evaluation draw, and inner replica each have a registered role and seed stream. The outer disorder unit is the independent unit for uncertainty.

Scale. $\mathbf{N} = (d, L, T, H, n, C, \dots)$ is a vector. One component is declared the finite-size coordinate and the remaining components follow a registered path; the scalar word “size” alone is insufficient.

Hierarchical six-axis taxonomy



Calibrate on source conditions; report uncertainty and untouched holdout outcomes

Figure 1: The paper in one figure. A atlas coordinate selects a deformation; effective variables can then be calibrated at an anchor, scored on an untouched holdout, and assigned an outcome class. Taxonomy (what exists), protocol (what is tested), and evidence (what may be claimed) remain separate.

Observable and phenomenon. An observable records estimator, units, intensive/extensive character, ensemble, null, and validity range. A phenomenon is then a hypothesis about recovery, specialization, locality, multiplicity, symmetry breaking, crossover, criticality, glassiness, dynamical arrest, or collective alignment—never a synonym for a visually steep curve.

2.1 What is measured

An order parameter must retain meaning under deformation. We group observables by function rather than by domain (Table 2). In particular, $K_{\text{eff}}^{\text{macro}} = \exp H(Z)$ counts effective macrostates, $N_{\text{eff}}^{\text{corr}}$ discounts correlated components, and $r_{\text{eff}}^{\text{spec}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ is a spectral effective rank. They are not interchangeable effective numbers.

2.2 Outcomes and PhaseCards

An edge has one of 13 registered outcomes: preserved, renormalized, rounded, split, merged, new regime, hysteretic, path-dependent, statistical–computational separation, semantic failure, censored without a crossing, unresolved, or not comparable. These outcomes are hypotheses adjudicated by diagnostics, not labels assigned from visual smoothness.

The minimal experimental record is

$$\mathcal{P} = (D, S, \delta, \omega, \mathbf{N}, \mathbf{g}, \mathbf{m}, \mathbf{R}, H, T, F, \mathbf{E}, O), \quad (1)$$

Axis	Transformer	Diffusion	RL / RLVR	Multi-agent
Data / environment	synthetic, grammar, web text	mixture, manifold, image data	bandit, MDP, verifier tasks	private evidence, task graph
Model / interaction	QKV, heads, MLP, norm, residual	score model, U-Net, DiT, locality	tabular, MLP, Transformer policy	topology, bandwidth, memory, roles
Objective / feedback	MSE, CE, causal, SFT, preference	ϵ , x_0 , v , flow, guidance	gold, proxy, process reward	individual, team, competitive reward
Dynamics / optimizer	SGD-M, AdamW, Muon, SOAP, schedule	optimizer, noise schedule, solver	PPO, off-policy reuse, search	response and memory update
Lifecycle / population	pretrain, LoRA, pruning, RAG, rollouts	distillation, conditioning	SFT-RL, verifier calls	heterogeneous agents, dynamic topology

Table 1: Domain-specific coordinates within the deformation layer. The other atlas layers—state, disorder, scale, observable, and phenomenon—are recorded separately; no empty cell is implied to have been tested.

Family	Meaning	Representative probe	Required null
Recovery	latent truth recovered	counterfactual semantic or oracle probe	shuffled latent labels
Specialization	permutation symmetry broken	head, branch, strategy, or role assignment	permuted components
Locality	range of causal influence	matched site or message perturbation	random-site perturbation
Multiplicity	independent computational units	correlation participation	independent-replica null
Entropy / commitment	macrostate selection	strategy or trajectory entropy	token-level entropy alone
Replica structure	diversity of solutions	functional overlap distribution	seed-matched random solutions
Robustness	quality of the ordered state	OOD risk, gold reward, truth consensus	proxy-only performance
Resource response	marginal value of compute	matched FLOP, NFE, rollout, or message intervention	equal-resource control

Table 2: Order-parameter dictionary. A PhaseCard records the exact estimator, probe distribution, normalization, units, and null.

where D is domain, S the state object, δ deformation, ω disorder hierarchy, $\mathbf{N}$ scale vector and path, $\mathbf{g}$ controls, $\mathbf{m}$ observables, $\mathbf{R}$ resources, H holdout rule, T theory status, F fidelity, $\mathbf{E}$ the evidence vector, and O outcome. Full PhaseCards are machine-readable; representative cards appear in Section 6.

3 Theory frontier and the unresolved gap

The literature registry separates a bibliographic claim from an implementation and from evidence generated here. It records publication status, model class, state object, assumptions, predicted observables, and the atlas edge for which the source is relevant. The taxonomy is our organizational proposal, informed by reproducibility frameworks that make model and dataset assumptions explicit [15, 21, 25]. It is not presented as an established consensus.

For attention, Cui et al.’s published tied low-rank model exhibits a positional-semantic transition [6], and attention-indexed models extend Bayes predictions to full-width matrices [4]. The open

Anchor setting	Established result	Restrictive assumption	Deformation tested here
Tied attention	positional–semantic boundary	low rank, attention only	trainable block and module screens
Attention-indexed model	Bayes boundary	simplified data and loss	optimizer, objective, natural carrier
Oracle diffusion score	speciation and collapse regimes	oracle score	trajectory and learned-score validation
Proxy-reward model	Goodhart scaling	controlled verifier	KL–noise phase map
Mean-field consensus	field/coupling response	homogeneous agents	population and field holdouts

Table 3: Theory frontier and the assumption removed by the benchmark. The last column states a test, not a guarantee of successful continuation.

question is therefore not whether simplified attention can transition, but whether its boundary and order semantics survive standard blocks, MLPs, normalization, causal objectives, optimization, and natural carriers.

Diffusion theory distinguishes speciation from later memorized-point collapse in the published analysis of Biroli et al. [3]; DDPM and DiT provide the relevant learned endpoints [16, 23]. A recent unreviewed preprint reports nearby symmetry-breaking and nonlocality onsets [30], which we treat as a hypothesis rather than settled evidence. In RL, proxy optimization can reduce gold quality [14, 26], so entropy must be read jointly with gold reward and strategy support. Consensus dynamics have a classical foundation [8]; modern MARL and LLM debate add nonstationarity and heterogeneous language agents [9, 20]. The 2026 statistical-physics agent papers are explicitly recent preprints. De Nobili reports model-dependent crossovers dominated by intrinsic bias rather than evidence for a two-dimensional Ising transition; we use this as a cautionary hypothesis, not a settled frontier [1, 7].

4 Research questions and dossier map

The atlas is organized around seven questions rather than around a list of available simulators. RQ1 asks when a semantically validated order parameter survives a deformation. RQ2 asks whether a boundary sharpens, crosses, rounds, or becomes censored under finite-size scaling. RQ3 asks which modules or interactions causally support an ordered state. RQ4 asks how optimization and objective geometry alter the route to that state. RQ5 asks whether a numerical relation predicts an untouched size, control window, or realism edge. RQ6 asks which conclusions survive a natural-data or learned endpoint. RQ7 asks which failure mode—semantic loss, scale dependence, path dependence, or model misspecification—explains unsuccessful transport.

Each domain dossier uses the same four-part report: *observation* states a registered estimate; *interpretation* states the narrow physical reading supported by its null and evidence vector; *not established* prevents a nearby stronger claim; and *next falsifier* names the experiment that could change the status. The machine-readable question and implementation registries link each RQ to runnable PhaseCards while leaving known-but-unimplemented and unknown cells visible.

5 Experiment grammar and predictive phase continuation

Every run is generated from four typed objects. A *ScientificSpec* names the question, state object, deformation, phenomenon, theory status, fidelity, and observables. A *ProtocolSpec* fixes controls, outer and inner disorder, finite-size path, nulls, interventions, calibration, and holdouts. An *ExecutionSpec* fixes implementation, resources, seed streams, and software identity without changing

scientific identity. An *EvidenceVector* records semantic validity, replication, finite-size diagnostics, prospective prediction, intervention, and external validity.

The output of an estimator is not a bare float but

$$\hat{O} = (\bar{O}, [O_-, O_+], q, U, n_{\text{outer}}, n_{\text{inner}}, \mathcal{S}_{\text{outer}}, \text{units}), \quad (2)$$

where q is interval level and U names the uncertainty method. The ensemble is part of the observable: checkpoint-to-checkpoint variability is a temporal statistic, whereas susceptibility and Binder cumulants used here are computed over independent terminal outer-seed samples.

5.1 Predictive phase continuation as one atlas protocol

Let $\mathcal{O}_{\text{cal}}$ be registered calibration observables and $\mathcal{O}_{\text{solv}}(\theta)$ a solvable surrogate. We estimate

$$\hat{\theta}_{\text{eff}} = \arg \min_{\theta} D\left(\mathcal{O}_{\text{cal}}^{\text{target}}, \mathcal{O}_{\text{solv}}(\theta)\right), \quad (3)$$

then predict an order curve, a boundary $\hat{g}_c$, transition diagnostics, and an intervention response. The bridge error is evaluated only on withheld quantities,

$$\mathcal{E}_{\text{bridge}} = D\left(\mathcal{O}_{\text{holdout}}^{\text{target}}, \mathcal{O}_{\text{solv}}(\hat{\theta}_{\text{eff}})\right). \quad (4)$$

Semantic retention $S_{\text{sem}} = I(m; Z)/H(Z)$ is a separate estimand: a small boundary error is meaningless if m no longer tracks the latent variable Z .

Baselines. Every predictive comparison includes a constant predictor, direct transport of the source boundary, a size-only or nearest-calibration predictor, the base surrogate, and the registered augmented surrogate. A diagonal plot without these controls cannot distinguish physical transport from simulator reparameterization.

Breakdown and repair. Two assumptions can interact through

$$\Delta_{ij}g_c = g_c^{ij} - g_c^i - g_c^j + g_c^0. \quad (5)$$

Holdout A is used only to diagnose a missing coordinate. Candidate macrovariables and their selection rule are frozen on a calibration split. Only Holdout B can support a repair claim. When an augmented surrogate was specified before either holdout, we call it a preregistered comparison, not a post-hoc repair.

Prediction target. A threshold estimate is recorded as crossed, left-censored, or right-censored. Only crossed estimates enter boundary-error summaries. Censored curves remain visible because the absence of a crossing is often the scientific result.

Attempt and invalidation semantics. A failed accelerator attempt and a CPU fallback are separate attempts under one scientific condition. Artifact identity records scientific, protocol, implementation, and execution hashes separately. If a metric definition, ensemble, split, or selection rule changes, the prior artifact is retained but marked invalidated for the affected claim; it is never silently reinterpreted.

6 Experimental and statistical protocol

Evidence strength follows the vector summarized in Table 4. The controlled predictive benchmark uses 16 independent outer-disorder seeds, four inner stochastic replicates, and 16 additional outer

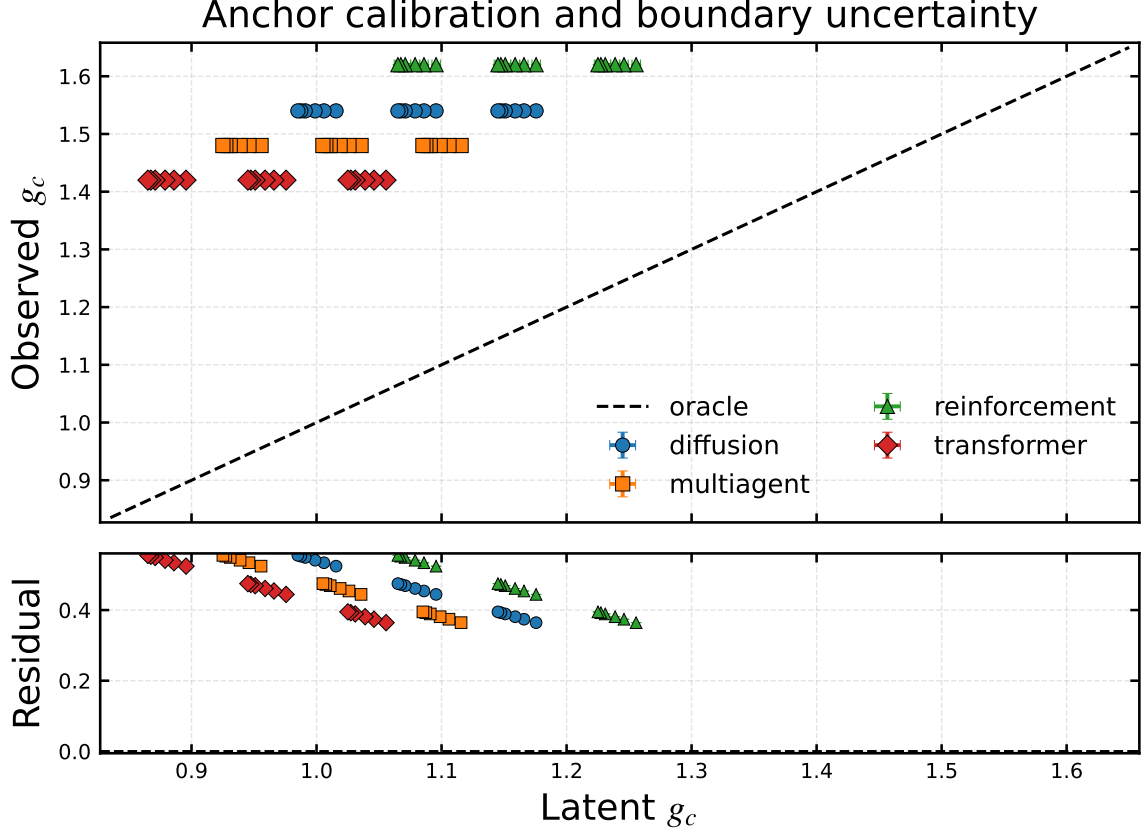


Figure 2: Anchor calibration before continuation. Observed pseudocritical controls are compared with registered latent references, with outer-seed 95% intervals on both axes and a residual panel. This is a calibration diagnostic, not held-out evidence for transport.

seeds near the frozen boundary window. Inner replicates are never counted as independent disorder. The broad Transformer taxonomy uses exactly five full-pipeline seeds per condition and Student- t intervals; it is a screen. Frozen confirmation uses twelve fresh outer seeds. These designs answer different questions and are never pooled.

Finite-size inference. Finite-size scaling is domain specific [2, 12, 13]. We therefore do not impose $N^{-1/2}$ or N^{-1} on width, trajectory resolution, horizon, and population as universal exponents. Candidate forms are empirical models whose parameters are fit below the largest size and selected without it; the untouched largest size is used only for prospective scoring. A phase label additionally requires order semantics, sharpening or crossing diagnostics, sensitivity to the minimum size and fitting window, and a registered alternative.

Intensive observables. For a vocabulary of size V , cross entropy is reported as $R_{\text{CE}} = \ell_{\text{CE}}/\log V$; the Brier risk is divided by $1 - 1/V$. We report R_{train} , R_{IID} , R_{OOD} , and $e_{\text{gen}} = R_{\text{IID}} - R_{\text{train}}$ separately. FLOPs, tokens, wall time, width, depth, context, and sample count remain resource coordinates. Causal module effects use raw risk differences and a bounded signed normalization, never division by a vanishing remaining-accuracy scale.

Accounting and preregistration. A condition is a unique PhaseCard; a trained model is one outer seed; an inner replicate is conditional stochasticity; an evaluation is not a new model. Metric schemas must match exactly across registered seeds: aggregation never takes the intersection of available keys. A boundary-profile bootstrap draws one outer-seed index vector and reuses it

Tier	Purpose	Sizes	Outer seeds	Allowed language
A	finite-size phase or boundary transport	registered path	12–32	phase structure, crossing/crossover, predictive boundary
B	frozen mechanism or endpoint comparison	≥ 3	≥ 5	shift, mechanism candidate, resolved/unresolved contrast
C	natural or open end-point	≥ 2	≥ 5	measurement transport and bounded external validity

Table 4: Convenient language classes derived from, but never replacing, the evidence vector. No fixed size count by itself upgrades an invalid order parameter or a non-independent holdout.

across all controls, preserving the paired disorder profile. Configuration hashes, corpus revisions and hashes, seed values, raw seed metrics, censor flags, cap-hit flags, tokens, estimated FLOPs, and elapsed time are stored before aggregation. Confirmation configurations were frozen after the exploratory audit in the machine-readable claim ledger [22].

7 Transformer flagship study

The Transformer is the flagship because all six realism axes can be varied while retaining a direct task-success probe. The study has two layers: a five-seed atlas that locates candidate effects and a twelve-seed frozen suite that tests only claims selected before the new seeds are observed.

7.1 Data and attention anchor

The controlled retrieval task contains a latent key–value relation and a positional shortcut. The primary task order is held-out answer accuracy; attention entropy, causal range, and representation overlap are secondary diagnostics. Figure 3 shows the Tier-A controlled anchor with raw disorder points, susceptibility, drift, and largest-size model comparison. Performance-derived $1 - R$ is retained as a learning diagnostic, not silently promoted to a thermodynamic order.

7.2 Feed-forward modules: from screen to matched confirmation

The atlas varies no MLP, linear, ReLU, GELU, GEGLU, and SwiGLU blocks across width and sample coefficient. Its completed five-seed aggregate shows that gated MLPs do not monotonically improve the selected attention-dominant endpoint. This is an MLP-dependent response, not yet evidence of a new phase. The frozen suite compares linear/GELU blocks with gated blocks whose hidden ratio is reduced from 4 to 8/3, matching feed-forward parameter count. It records full, attention-ablated, MLP-ablated, and jointly ablated $O(1)$ risks.

At the largest registered endpoint, the normalized task orders are 0.514 ± 0.007 (linear), 0.521 ± 0.007 (GELU), 0.433 ± 0.016 (GEGLU), and 0.439 ± 0.013 (SwiGLU). The corresponding bounded MLP effects are 0.254 ± 0.080 , 0.310 ± 0.092 , 0.187 ± 0.049 , and 0.183 ± 0.026 . These values are generated directly from the frozen 12-seed aggregate rather than transcribed by hand.

7.3 Optimizer and normalization

Optimizer names are grouped by geometry: scalar Euclidean (SGD-M), diagonal adaptive (AdamW), matrix orthogonalized (Muon), and matrix preconditioned (SOAP), with Lion and GaLore retained as exploratory extensions [5, 18, 19, 28, 31]. Learning rates for the frozen comparison are selected only on widths $d \leq 96$ and $\alpha \leq 2$ of the completed atlas, then fixed for $d \in \{64, 96, 128\}$ and $\alpha \in \{4, 8\}$. Endpoint risk, gradient heterogeneity, noise scale, relative

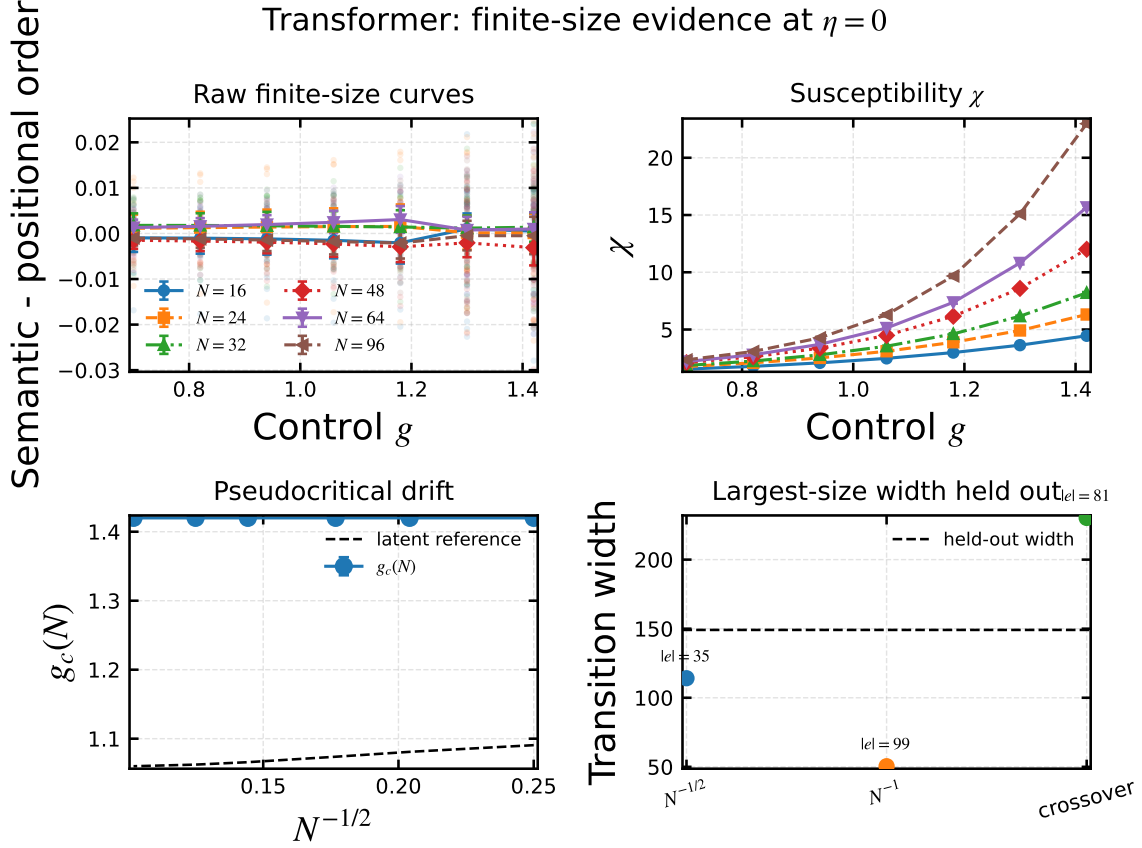


Figure 3: Transformer anchor finite-size evidence. Panels separate the raw order curve, response, pseudocritical drift, and held-out model comparison. Only explicit crossings are boundary estimates; endpoint-only curves are censored.

update, and time-to-order are reported jointly. Geometry correlations are not interpreted as causal without update/noise matching.

The held-out normalized risks are 0.608 ± 0.001 (SGD with momentum), 0.546 ± 0.004 (AdamW), 0.539 ± 0.003 (Muon), and 0.830 ± 0.006 (SOAP), each with its registered 95% Student- t interval.

7.4 Objectives, residuals, and scaling paths

Objective interpolation and residual/normalization grids are Tier-B screens. They are useful when a boundary, task order, calibration, or time-to-order changes; overlapping terminal losses are not counted as phase evidence. Width, depth, context, and data are kept as distinct coordinates [17, 29]. The frozen data scaling suite restricts $d \leq 128$, $\alpha \leq 2$, and $\gamma \in \{1, 3/2, 2\}$ with a cap above the largest requested αd^γ , eliminating the cap artifact found in the atlas.

At the largest common endpoint, the normalized risks are 1.019 ± 0.028 , 0.459 ± 0.005 , and 0.434 ± 0.004 for $\gamma = 1, 3/2, 2$, respectively.

7.5 Synthetic-to-natural bridge

The bridge ladder distinguishes latent synthetic tasks, natural-carrier injection, controlled stories, and web corpora. TinyStories, SimpleStories, FineWeb-Edu, and Dolma are version pinned [10, 11, 24, 27]. Raw corpus experiments report bits per byte and train/IID/OOD risk on non-overlapping byte ranges. They test measurement transport, not a universal phase boundary. Only natural-carrier injection retains a known semantic label suitable for a boundary test.

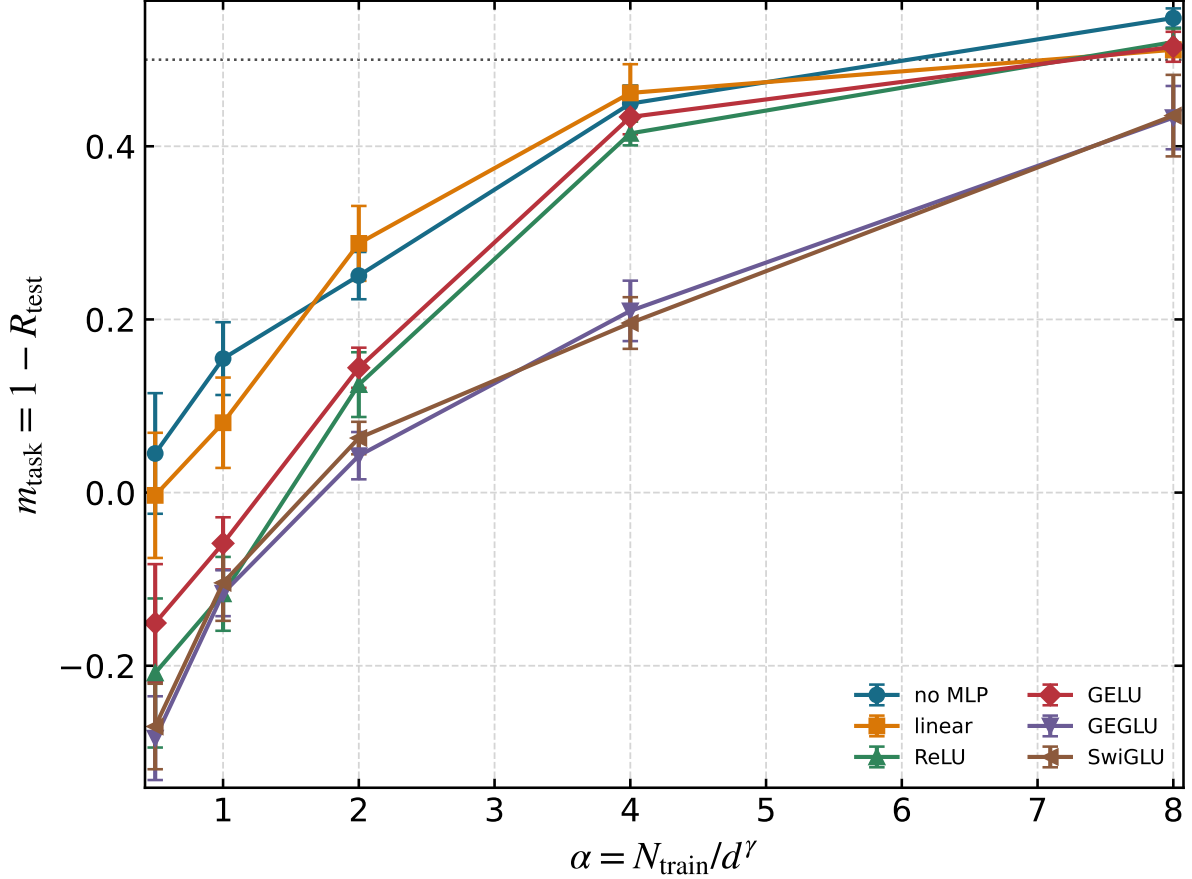


Figure 4: Exploratory MLP-dependent task order. Points are five-seed means with 95% Student- t intervals. The plot identifies hypotheses for frozen confirmation; it does not by itself establish phase splitting.

For the byte-disjoint evaluation protocol, the endpoint bpb values are 2.670 ± 0.019 (TinyStories), 2.815 ± 0.023 (SimpleStories), 3.427 ± 0.025 (FineWeb-Edu), and 3.260 ± 0.022 (Dolma).

8 Cross-domain validation

The following domains are canonical validation cases, not coequal mechanism studies and not evidence for shared equilibrium exponents.

8.1 Diffusion trajectories

The control is reverse time and the secondary coordinate controls locality. We track speciation, nonlocality, trajectory commitment, and proximity to memorized samples. The oracle anchor separates speciation from later collapse [3]; the holdout asks whether their ordering survives a deformed score trajectory.

8.2 Reinforcement learning

RL is organized by KL pressure β_{KL} and verifier noise ϵ_{ver} . Gold reward, strategy support, and Goodhart gap are joint observables; entropy alone cannot distinguish productive specialization from collapse [14]. The two-dimensional map tests whether an oracle-calibrated critical region predicts the holdout feedback dynamics.

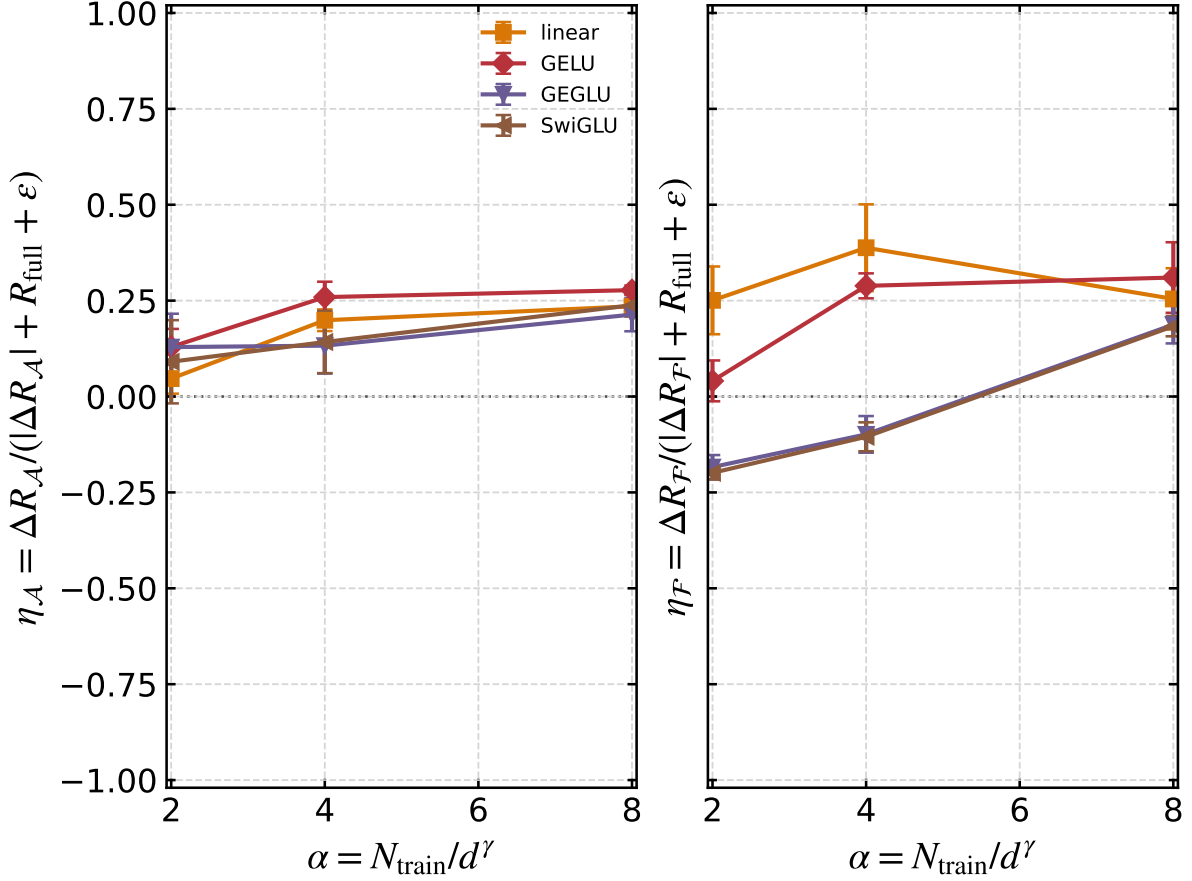


Figure 5: Bounded causal module effects from matched risk interventions. The raw risk difference is normalized by a non-vanishing risk scale, and therefore remains $O(1)$. Legacy ratios with vanishing denominators are excluded.

8.3 Multi-agent populations

Intrinsic evidence h and social coupling J are separate controls. A global flip prevents consensus from being mistaken for truth, and population size A remains explicit. The heatmap distinguishes field-driven alignment, interaction-driven consensus, fragmentation, and error amplification.

9 Synthesis: survival, breakdown, and intervention

9.1 Boundary transport against alternatives

Across the controlled benchmark, the base surrogate has median absolute boundary error 0.000, and the registered augmented surrogate has error 0.000. The relative change 99.8% is descriptive unless the augmented model also beats the constant, source-boundary, size-only, and nearest-calibration baselines on Holdout B. The mean anchor residual is 0.463. Prediction and benchmark intervals propagate the outer-seed boundary intervals through 256 deterministic parametric-bootstrap refits; they are not standard errors computed across heterogeneous holdout conditions.

9.2 What breaks and what can be repaired

The claim ledger distinguishes a preregistered augmented surrogate from a macrovariable selected after failure. Holdout A can diagnose that the base model omits specialization, nonlocality, strategy support, or intrinsic field. Selection occurs on calibration data; improvement on disjoint Holdout B is the repair test. The same points are never used both to invent and validate a macrovariable.

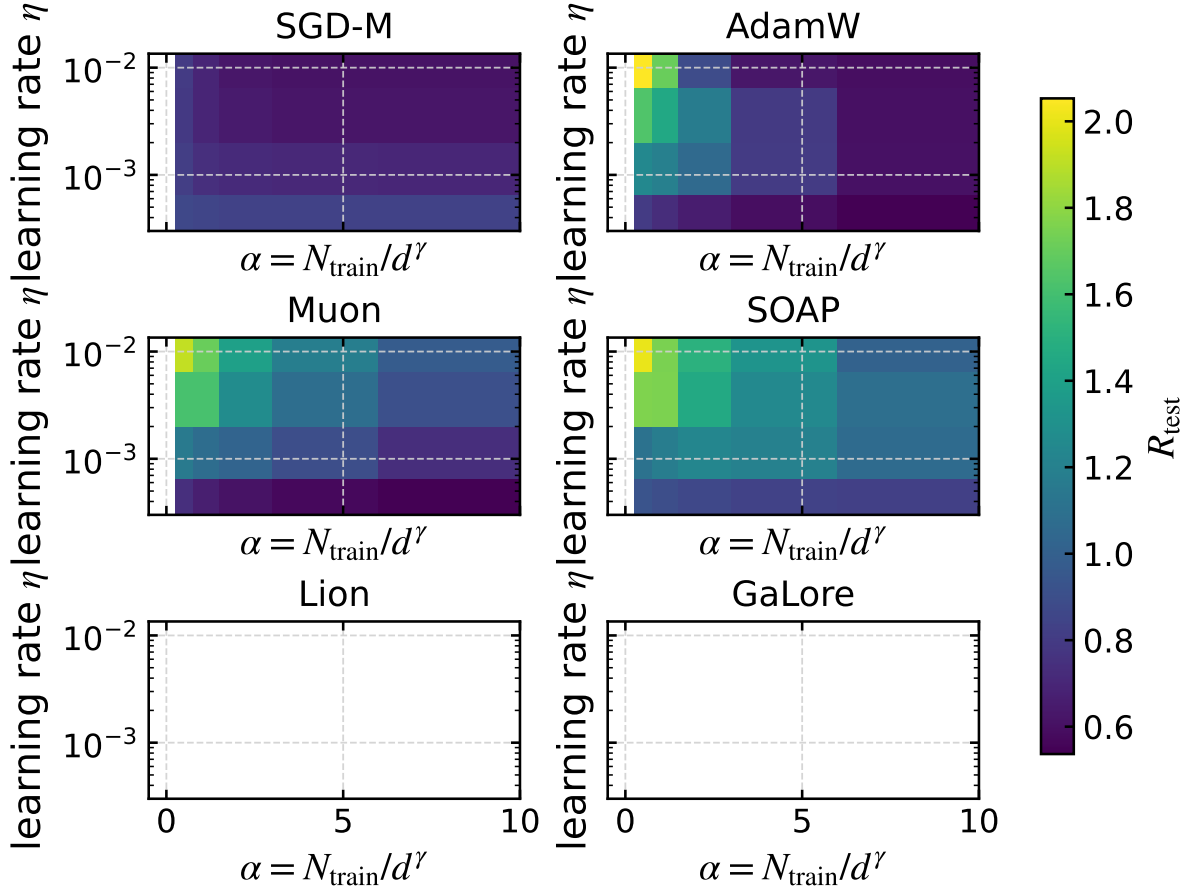


Figure 6: Optimizer response surfaces over learning rate and sample coefficient. Calibration and evaluation slices are marked explicitly; final risk is not used to tune the same point being reported.

9.3 Critical-window intervention

The controlled simulator compares always-on, always-off, random-window, matched-compute, and predicted-window policies. The predicted allocation removes 57.1% of uniform intervention compute while retaining 87.2% of its modeled quality gain; the mean gain over a random matched-compute window is 0.019. These values validate the benchmark logic, not wall-clock savings for a real training stack: the intervention benefit is generated by the registered controlled response model.

9.4 Claim ledger

10 Supplementary exploratory atlas

The exploratory atlas spans MLP, residual/normalization, objective, optimizer, scaling, and natural-data families. Figures 13, 17, 18, and 19 use the five-seed screen; Figures 12, 16, and 20 use the twelve-seed frozen confirmation. Every numerical panel reports 95% Student- t uncertainty. Screen plots locate effects and negative results; only hypotheses listed in the frozen claim ledger receive fresh confirmation.

The atlas label does not erase epistemic differences. The broad screen and confirmation runs are **S trainable synthetic** and predominantly **E empirical**; natural byte carriers change the data endpoint but do not constitute pretrained-model evidence. Phase observables in newly generated artifacts use independent terminal seeds. Older temporal proxies are retained only as dynamics observables and are invalidated for disorder-phase claims.

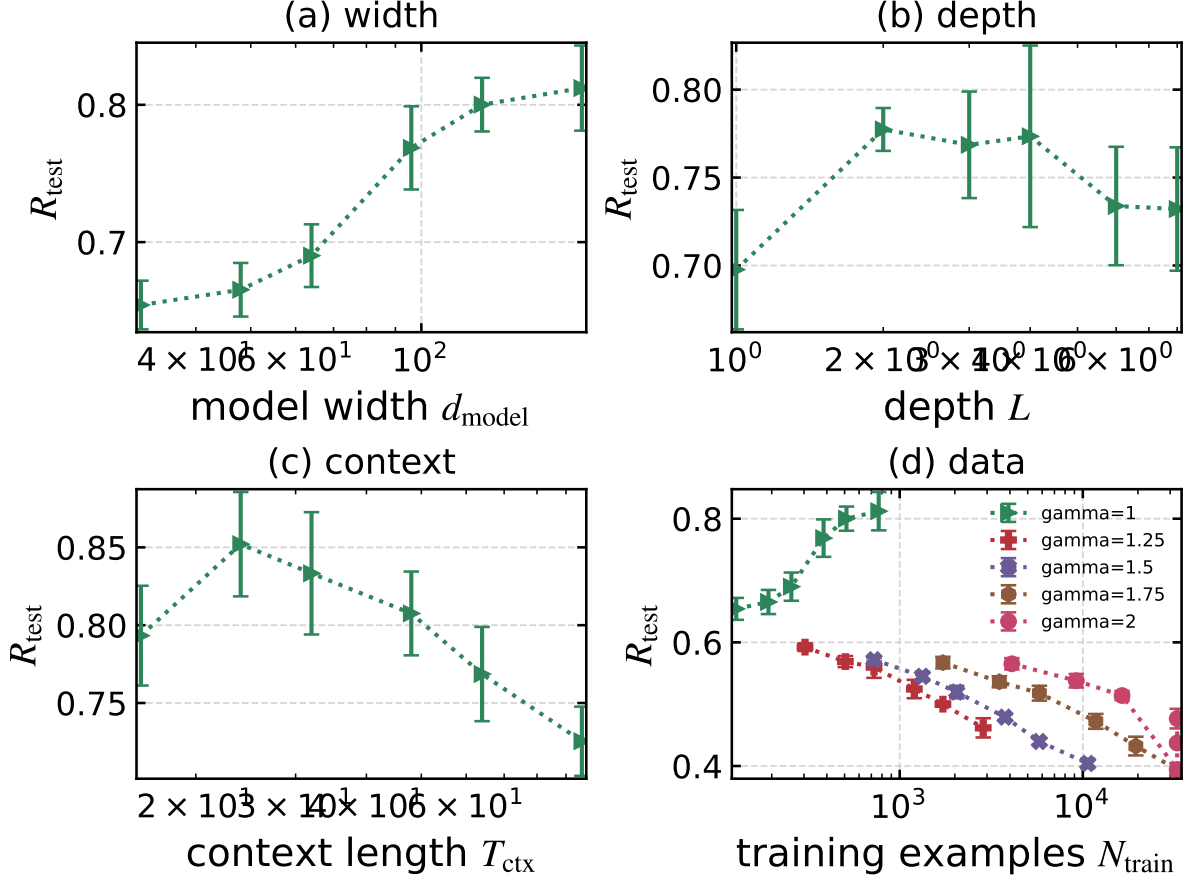


Figure 7: Scaling paths as separate physical small multiples: width d , depth L , context T , and sample count n in the five-seed screen. The frozen confirmation tests the three data exponents separately without imposing a common parameter-count coordinate.

11 Limitations and scope

The strongest evidence in this release is a controlled numerical benchmark, not natural language, natural image, open RLVR, or open-weight LLM-agent external validity. The Transformer atlas trains small decoder models from scratch. Natural corpora test a byte-level measurement protocol on disjoint ranges; they do not establish that a synthetic phase boundary transfers to web text.

The predictive benchmark is deliberately stylized. Its critical-window benefit follows a registered simulator response and is not an observed wall-clock training gain. Its augmented surrogate is a prospective comparison only where the extra coordinate was frozen before the evaluation split.

Most Transformer coordinates are Tier-B screens. Five seeds quantify full-pipeline variability but cannot prove an asymptotic universality class. Performance-derived $1 - R$ is not a semantic order unless a latent task probe supports that interpretation. Threshold curves that never cross are censored, not converted into endpoint boundaries.

Lifecycle, sparse/MoE feed-forward blocks, low-rank adaptation, preference/RL objectives, open checkpoints, and agent populations remain explicit open cells. Recent 2026 multi-agent and diffusion references are unreviewed preprints and are not used as sole support for a claim.

Temporal fluctuation proxies in a single training trajectory do not estimate disorder susceptibility or a Binder cumulant. The revised schema names these quantities separately and requires independent terminal samples for phase statistics, but legacy trajectories cannot retroactively supply that ensemble. Likewise, phenomenological generators are method-validation benchmarks rather than empirical evidence about trainable neural systems.

Finally, the confirmation suite was designed after auditing the broad atlas. Its fresh seeds

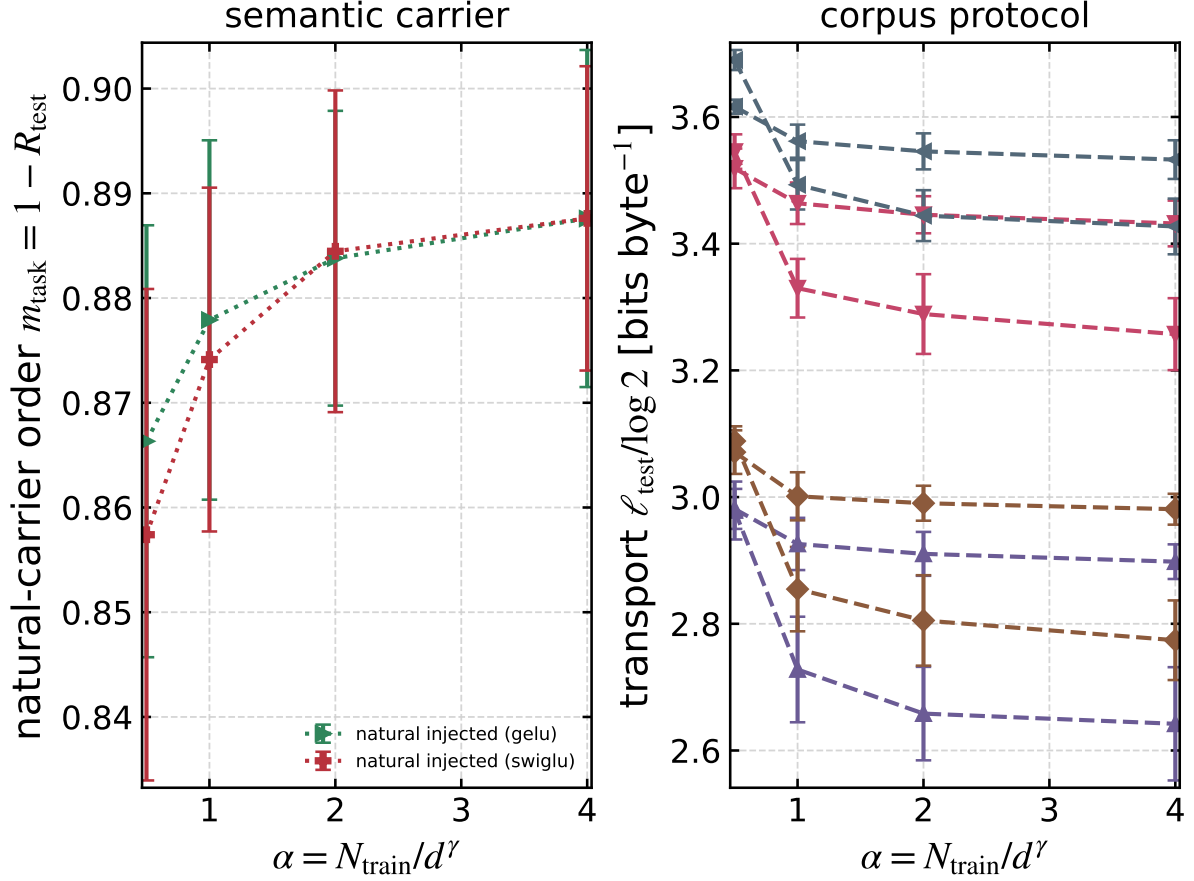


Figure 8: Natural-data bridge. Semantic order is shown only where latent truth is retained; raw corpora are compared in tokenizer-independent bits per byte. This five-seed bridge screen prevents protocol transport from being mistaken for boundary transport; twelve-seed raw-corpus confirmation is reported in Figure 16 and the generated endpoint values below.

protect the numerical comparison, but the hypotheses are second-stage confirmations rather than an independent replication by another implementation.

12 Conclusion

The atlas turns statistical-mechanics language into an auditable numerical practice. Its seven layers state what is studied; the PhaseCard states the ensemble, scale path, observable, alternative, intervention, and holdout; and the evidence vector states what may be claimed. Predictive phase continuation then turns selected atlas edges into prospective transport tests. Failure is informative: a boundary can shift, round, become censored, lose semantic meaning, or expose model misspecification without being forced into one universal story.

The Transformer study provides the deepest module, optimizer, scaling, and data tests. Diffusion, RL, and multi-agent systems show how the same evidence grammar can be used for trajectory, feedback, and population dynamics while retaining domain-specific observables. The central deliverable is therefore not twenty unrelated plots or a claim that modern ML shares one thermodynamic limit. It is an extensible map from questions to implementations, estimates, outcomes, claims, invalidations, and next falsifiers, with every untested cell remaining visible.

Diffusion: finite-size evidence at $\eta = 0$

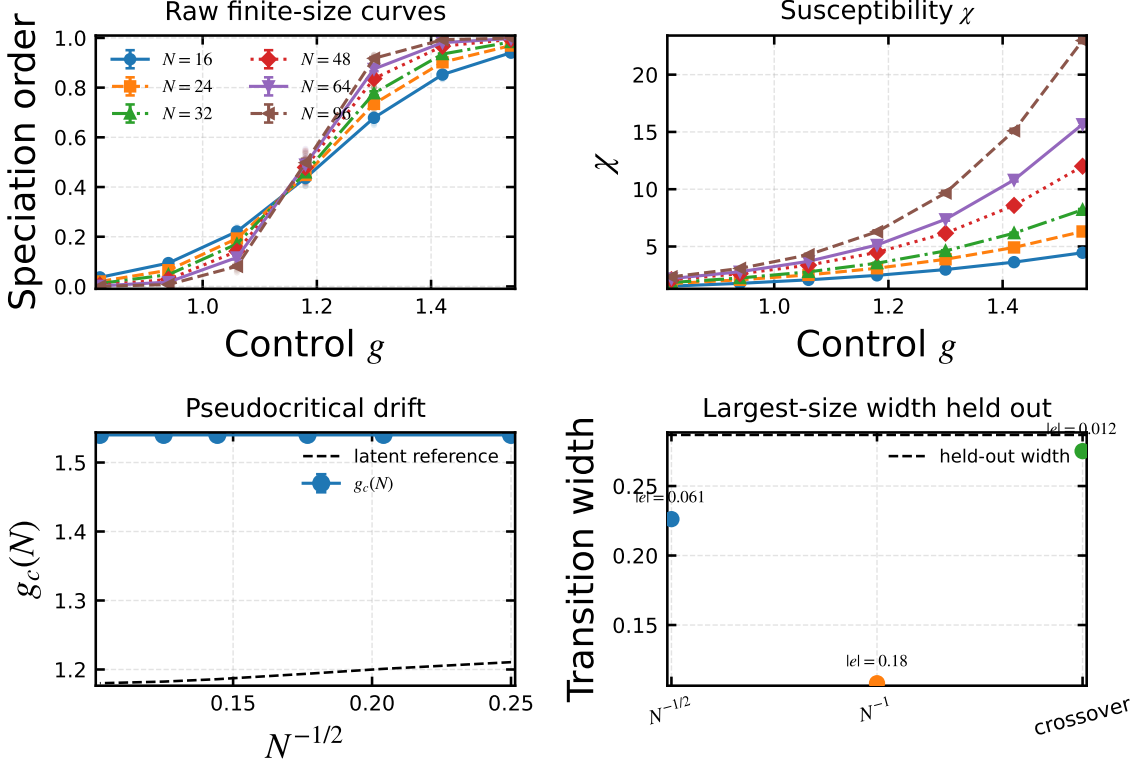


Figure 9: Diffusion finite-size validation. Raw order, response, pseudocritical drift, and largest-size predictive checks are shown separately.

A Reproducibility contract and representative PhaseCards

Repository code contains no site path. Production adapters provide repository, manifest, output, data, and Python locations through environment variables. Numerical production is rejected unless the Slurm partition begins with `spark_`. Each task writes atomically; strict aggregation requires the complete manifest seed set, finite required metrics, compatible trajectory checkpoints, and recorded censor and cap-hit status. The completed taxonomy screen contains 8,700 runs over 1,740 conditions. The frozen follow-up adds 1,632 runs over 136 conditions with 12 untouched seeds per condition.

Every publication figure is generated from a strict aggregate. Figures use a white background, physical size 6.4×4.8 inches, 12-point base type, larger mathematical axis labels, inward ticks, dashed grids, and fixed line, marker, and color order. PDF generation checks that every required figure and result-macro file is older than the compiled manuscript PDF.

B Negative-result taxonomy

C Artifact audit

The audit distinguishes source completeness from scientific validity. The completed exploratory tensor contains 8,700 trained models in 1,740 conditions, with five full-pipeline seeds per condition. A strict task count does not upgrade a performance-derived order, a censored threshold, a tuned endpoint, or an overlapping corpus split.

RL holdout landscape, $N = 96$

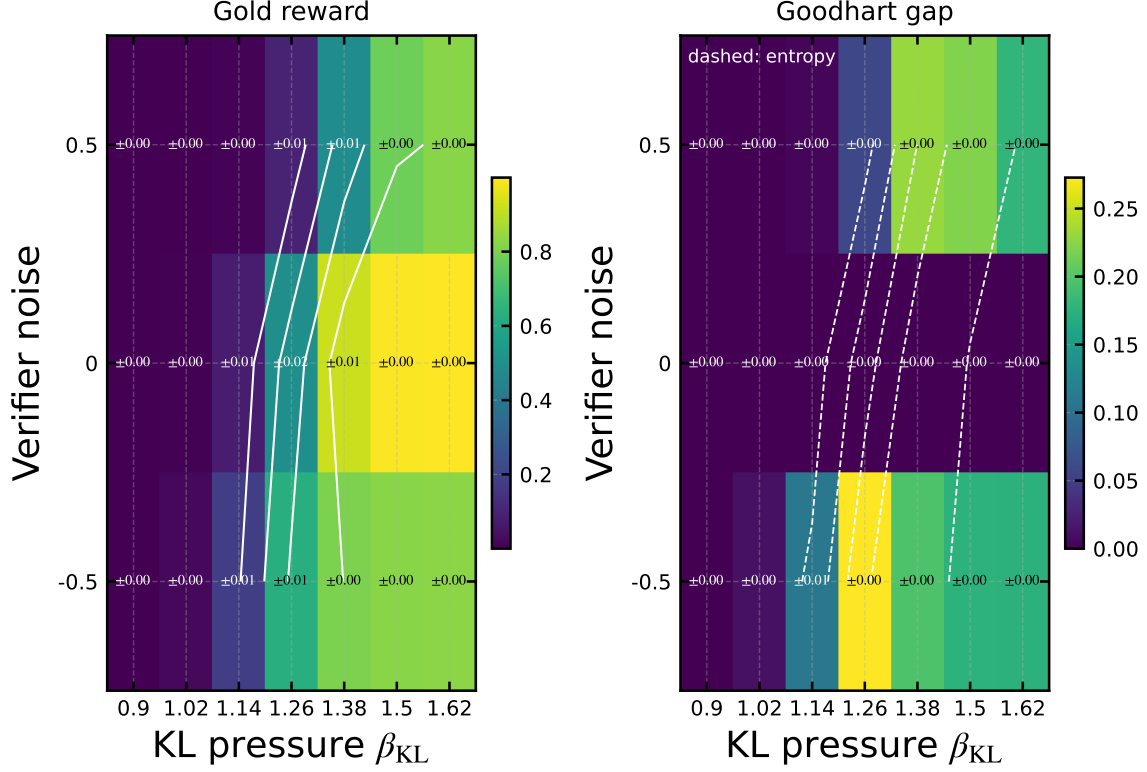


Figure 10: RL validation in the $(\beta_{\text{KL}}, \epsilon_{\text{ver}})$ plane. Color encodes gold reward; contours and companion diagnostics show strategy support and Goodhart behavior with uncertainty inherited from outer seeds.

Legacy artifacts are retained but not relabeled with statistics they did not record. The current invalidation ledger additionally prevents superseded metric or split definitions from supporting current claims. The audit found and corrected four material issues: an MLP causal ratio divided by a vanishing scale; threshold endpoints counted as crossings; optimizer learning rates selected on the endpoint being reported; and a quadratic sample path limited by `train_cap`. The confirmation suite was frozen specifically around these failures.

The revised artifact contract also distinguishes temporal trajectory variance from outer-disorder susceptibility and records failed accelerator and fallback runs as separate attempts. The full JSON stores raw seed values, between-seed uncertainty, train/IID/OD risks, trajectory checkpoints, resource accounting, dataset revision and hash, split ranges, crossing status, and cap-hit status. Aggregation fails on missing or non-finite required quantities.

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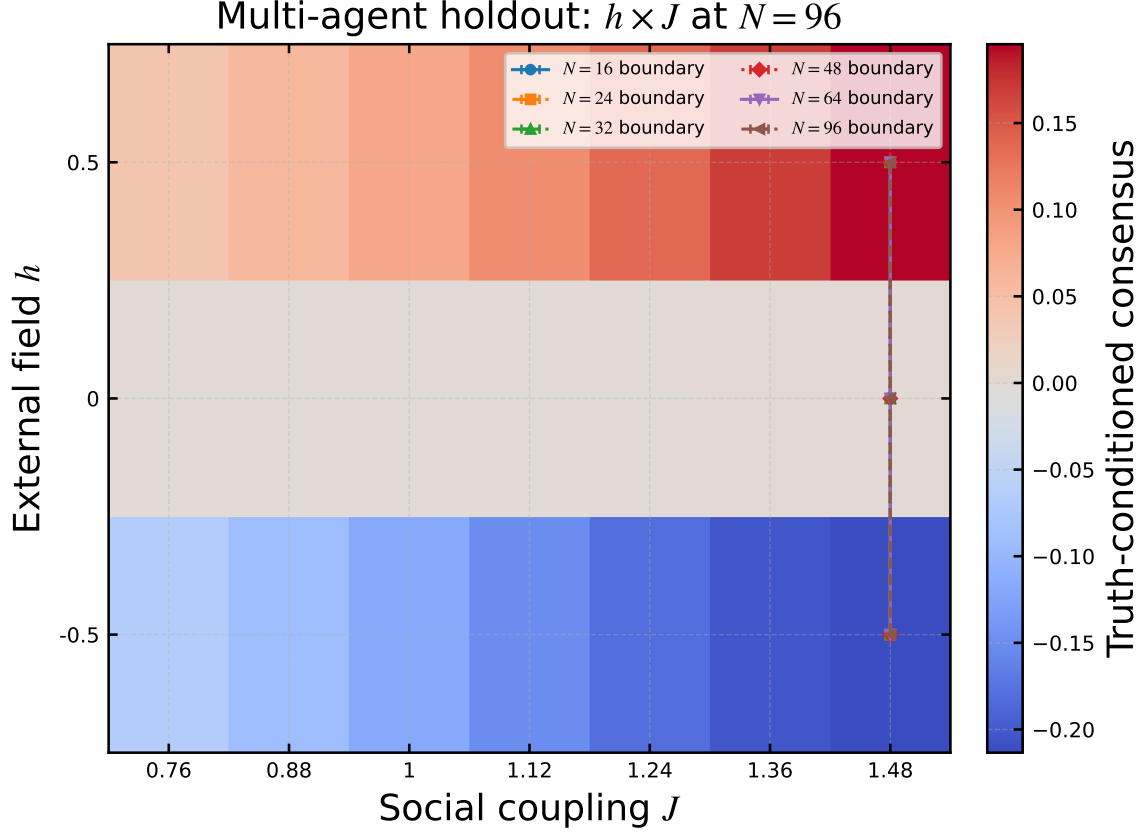


Figure 11: Multi-agent validation in the (h, J) plane with finite-population boundary overlays. The order is truth-conditioned consensus, not agreement alone.

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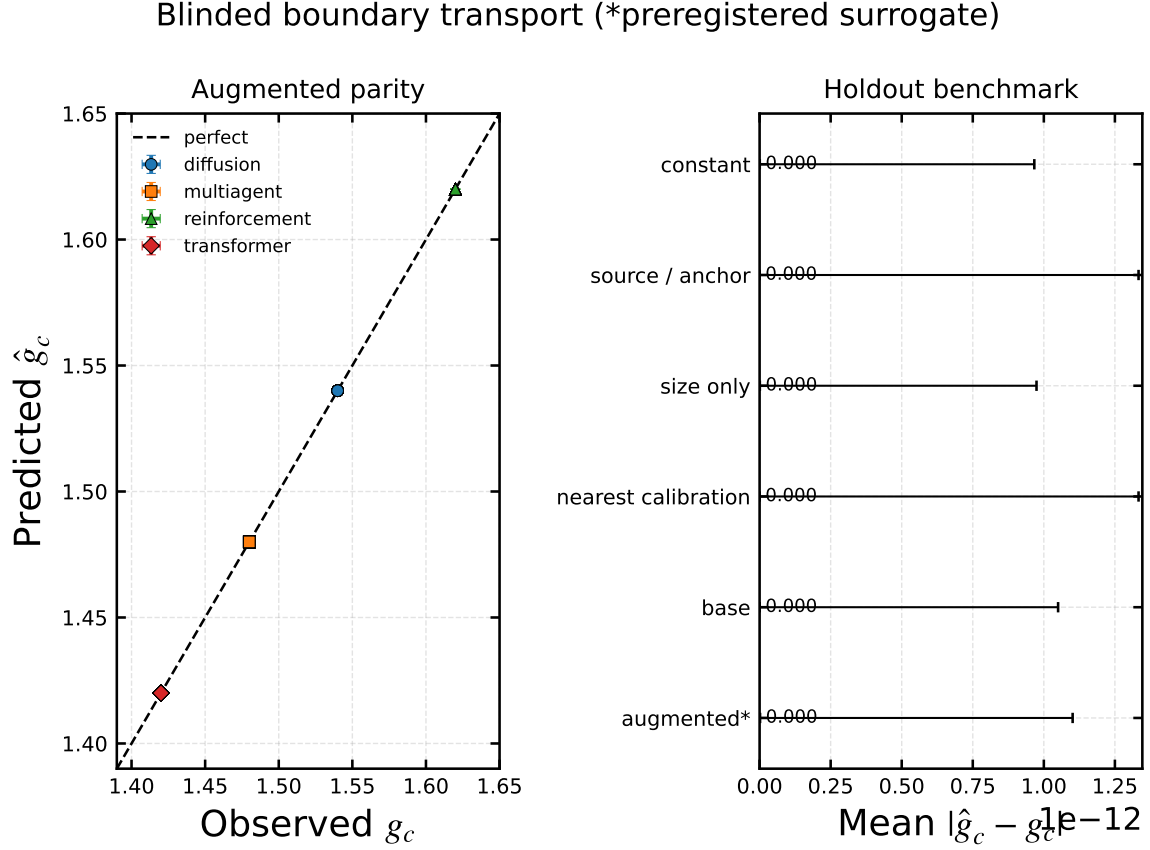


Figure 12: Held-out boundary transport and baseline errors. Marker shape identifies domain and holdout type; the companion panel compares absolute error against nontrivial transport baselines. Both axes and benchmark bars carry the propagated 95% intervals. A near-diagonal scatter alone is not the result.

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Deterministic split; no model selection or repair on Holdout A

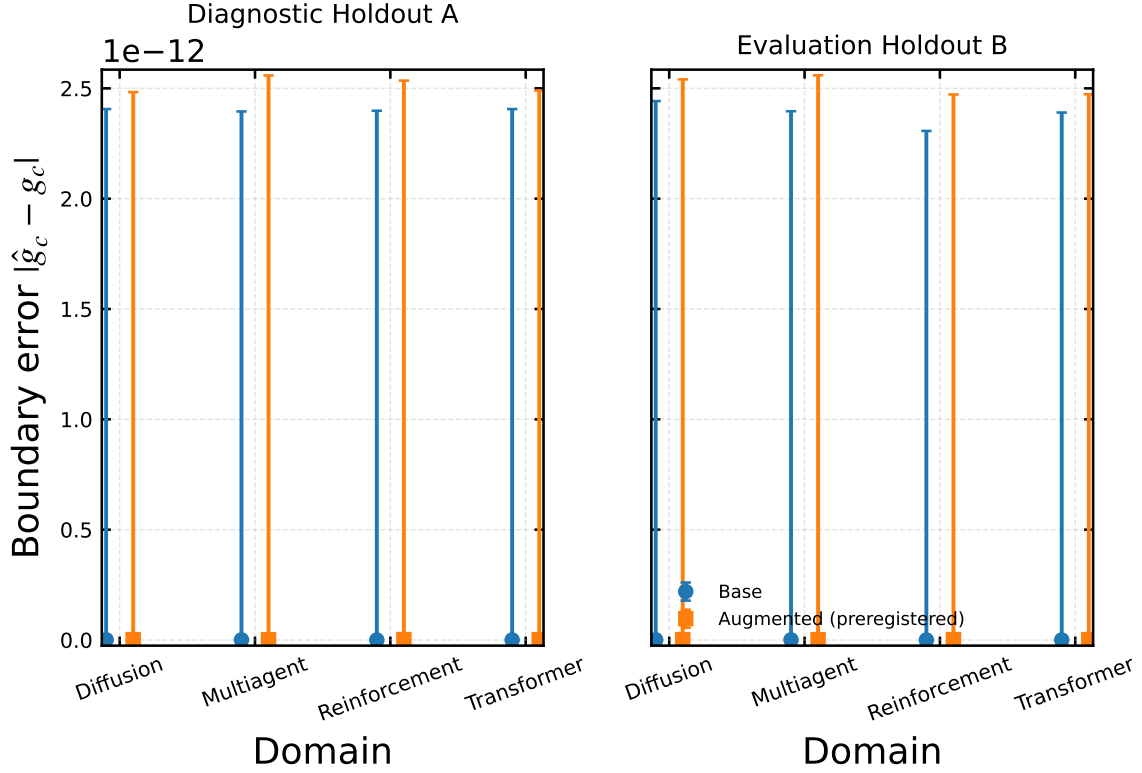


Figure 13: Breakdown and prospective augmentation. Holdout A is diagnostic; Holdout B is evaluative. Domain labels state the additional registered macrovariable, making the source of any improvement inspectable.

Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Oriol Vinyals, Jack Rae, and Laurent Sifre. Training compute-optimal large language models. *Advances in Neural Information Processing Systems*, 35, 2022.

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Domain	Base error	Augmented error	Modeled compute saving
Diffusion	0.000	0.000	57.1%
Multiagent	0.000	0.000	57.1%
Reinforcement	0.000	0.000	57.1%
Transformer	0.000	0.000	57.1%

Table 5: Generated controlled results. Boundary rows exclude censored thresholds, and the intervention column is explicitly simulator-level.

ID	Claim	Alternative	Holdout / artifact	Tier
C1	boundary transport	simple transport baselines	four-domain evaluation split / Fig. 12	A
C2	MLP module response	parameter count or terminal loss	matched fresh seeds / Figs. 4–5	B
C3	optimizer contrast	LR/update mismatch	frozen LR evaluation / Fig. 6	B
C4	scaling and corpus transport	cap or split overlap	uncapped and byte-disjoint suites / Figs. 7–8	B/C
C5	cross-domain grammar	Transformer-only validity	diffusion/RL/agent holdouts / Figs. 9–11	A

Table 6: Main claim ledger. The machine-readable version also stores anchor, primary order, falsifier, status, and artifact path.

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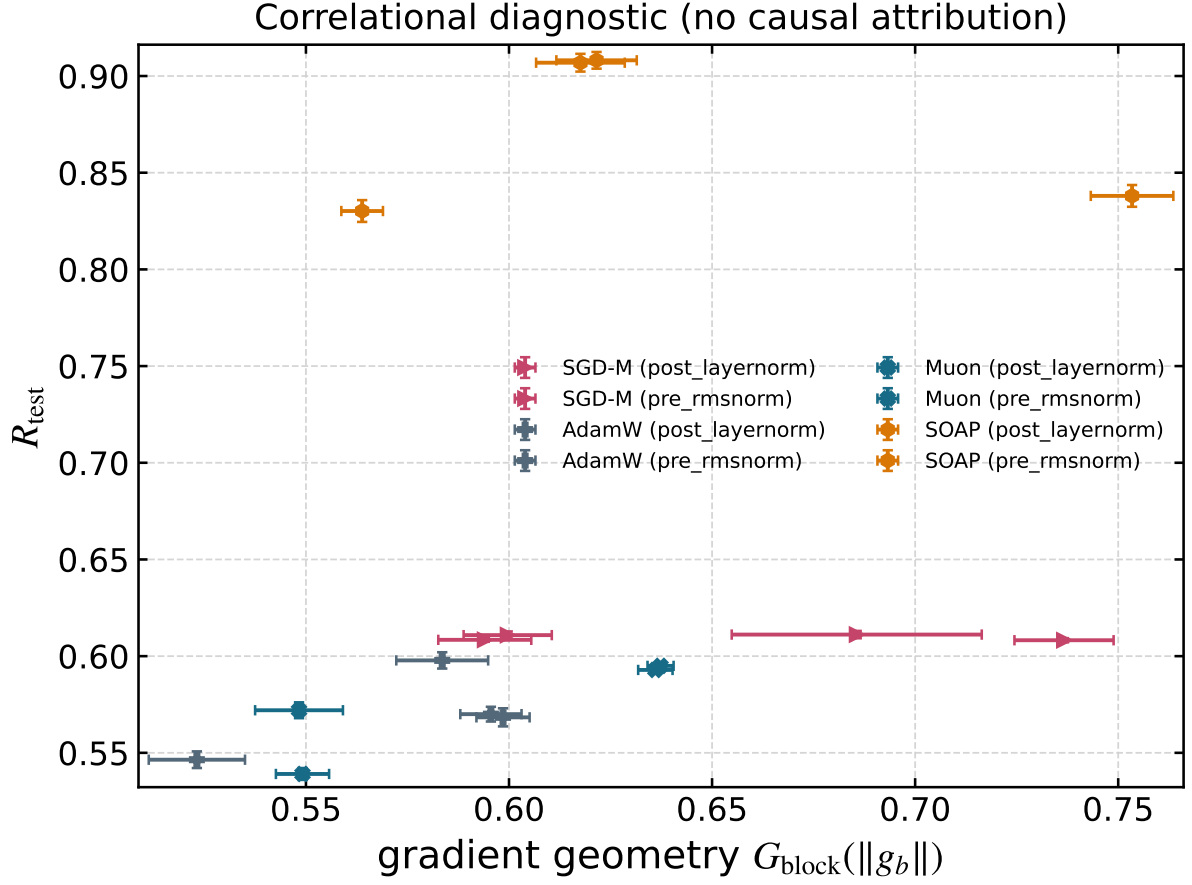


Figure 14: Gradient-block heterogeneity in the frozen optimizer comparison across three widths. Association with performance remains descriptive unless update and noise scales are matched.

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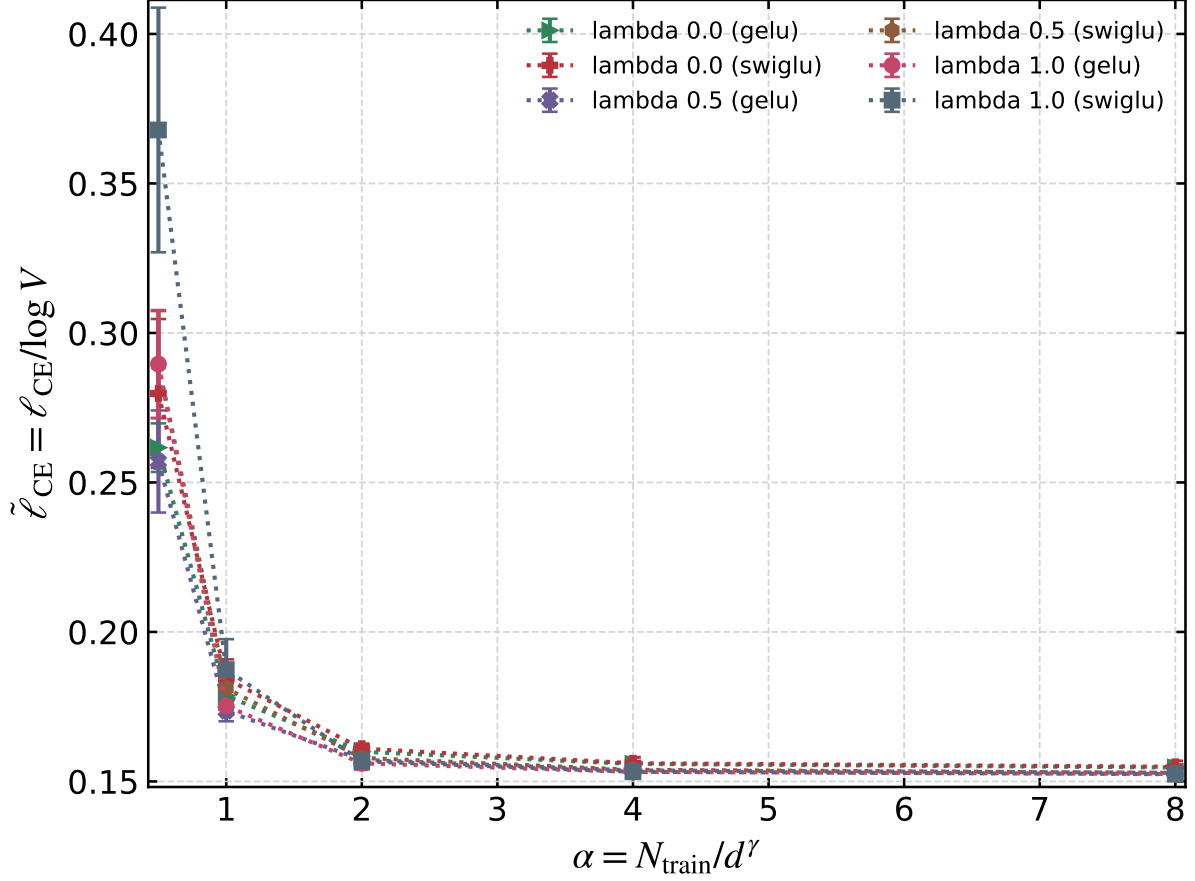


Figure 15: Objective interpolation screen. Overlapping terminal loss is not counted as a phase result.

Field	Predictive anchor	MLP confirmation	Optimizer confirmation	Natural endpoint
Level	representation	representation	learning dynamics	learning
Deformation	hidden domain rule	activation/gating	update geometry	corpus
Outer seeds	16+16	12 fresh	12 fresh	12 fresh
Primary order	semantic order	task success	R_{IID}	bits/byte
Alternative	transport baselines	parameter count	LR/update mismatch	split overlap
Holdout	hidden variant	matched modules	frozen LR slice	disjoint bytes
Tier	A	B	B	C

Table 7: Representative PhaseCards. Full cards and hashes are stored in the experiment configurations and manifests.

Outcome	Operational evidence	Allowed interpretation
No crossing	order remains on one side of threshold	censored threshold, not a boundary
Rounding	response width fails to sharpen prospectively	crossover on tested sizes
Semantic failure	probe loses mutual information with latent truth	abandon or replace the order
Optimization failure	information/order exists but training does not reach it	computational–statistical separation candidate
Unresolved contrast	interval and matched controls do not separate methods	no ranking
Cap limited	requested sample path hits storage/training cap	exclude from scaling fit

Table 8: Negative results are recorded outcomes, not omitted cells.

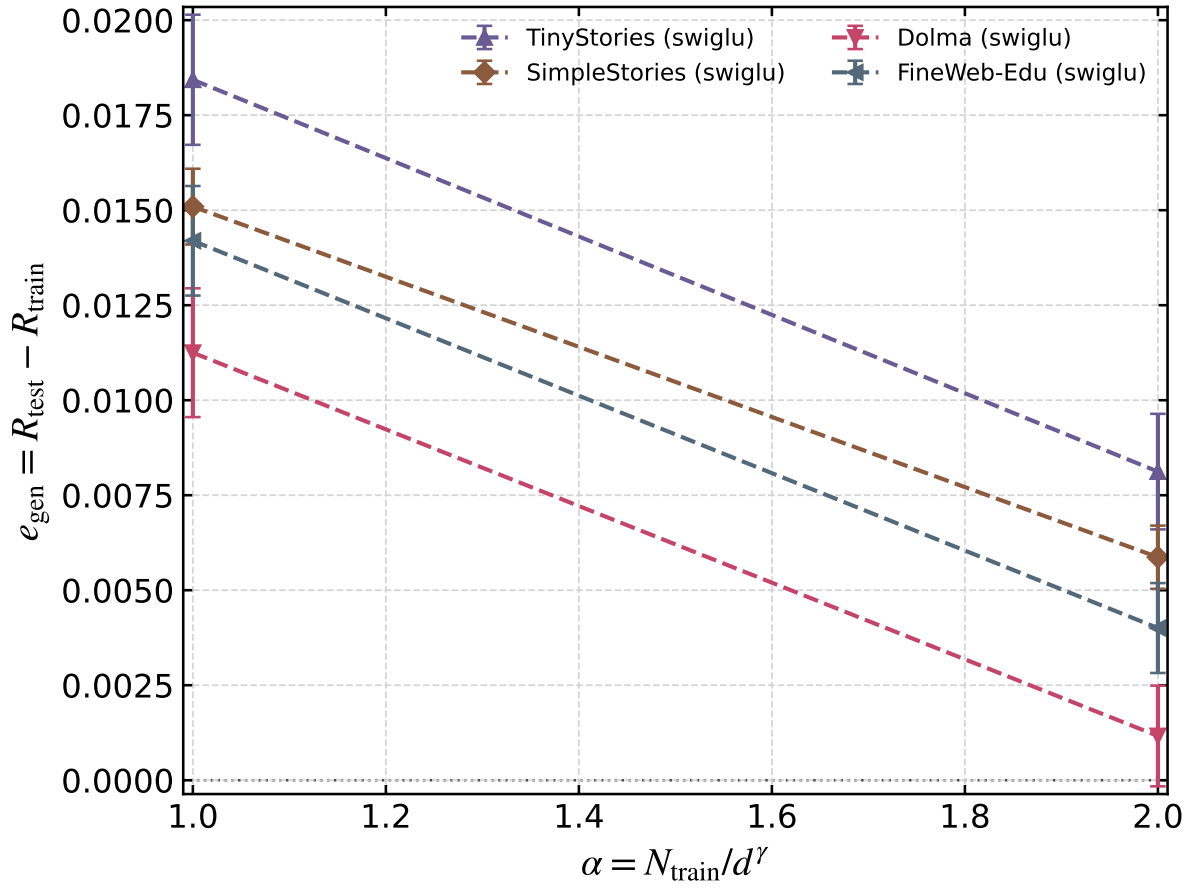


Figure 16: Train–test gaps on natural endpoints. Confirmation uses disjoint byte ranges and reports cross-corpus OOD risk separately.

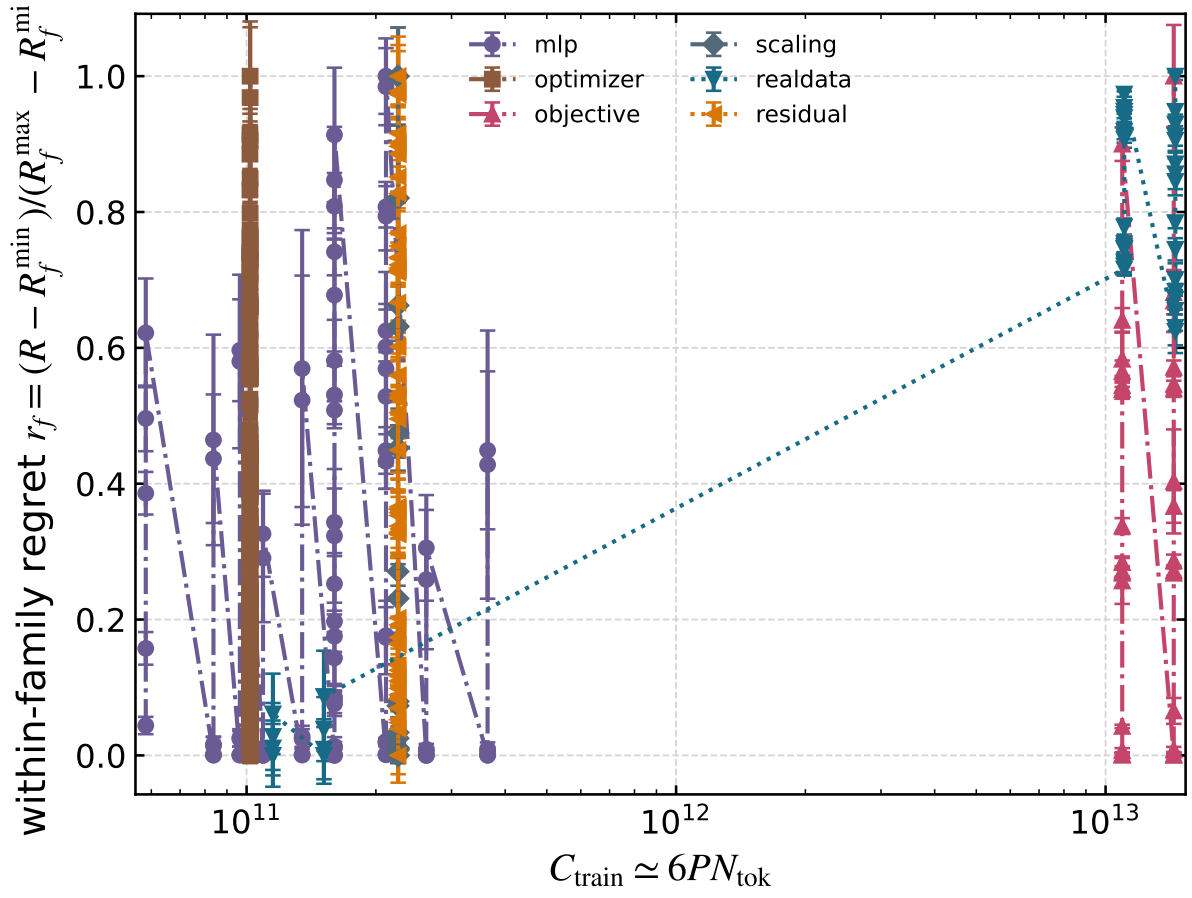


Figure 17: Within-family normalized regret versus estimated training compute. Raw risks from different tasks are never overlaid as if commensurate.

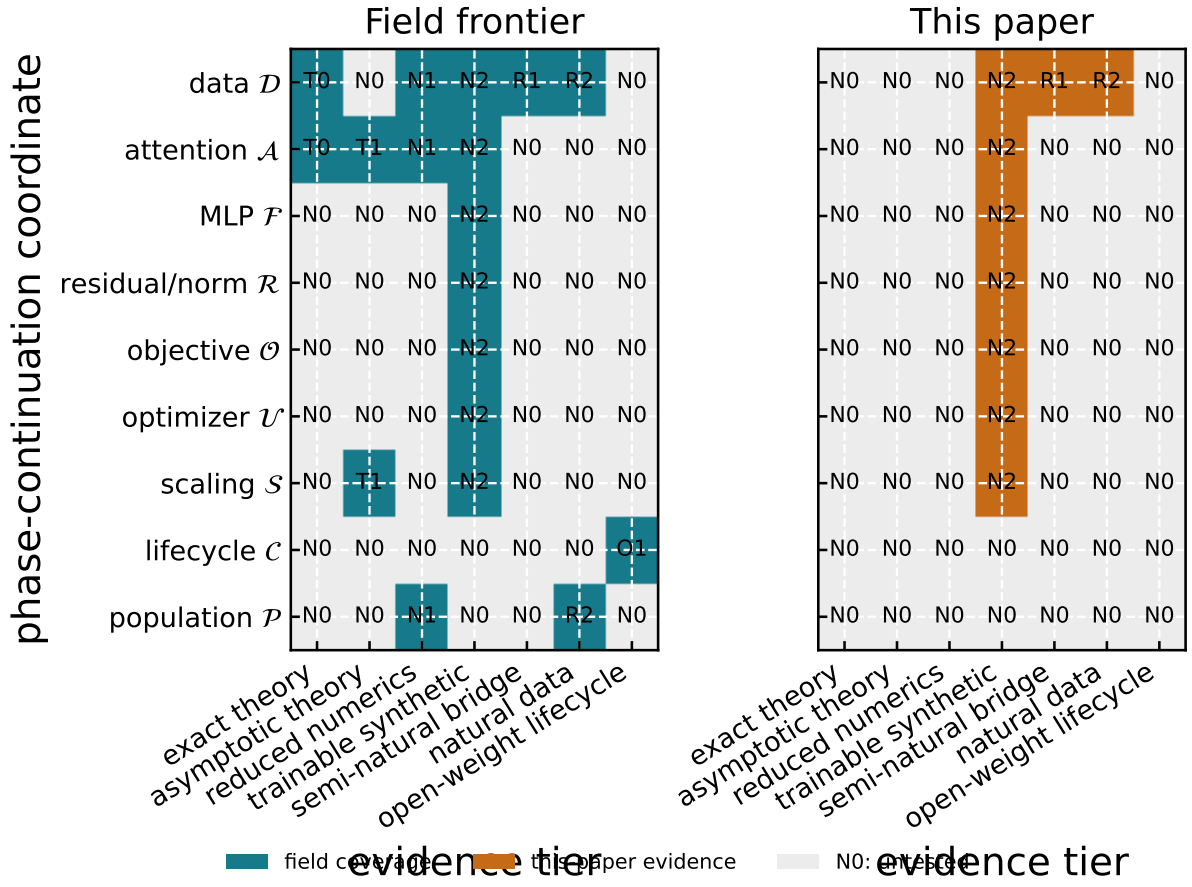


Figure 18: Field frontier and this-paper evidence are shown as separate matrices. An absence of theory is not confused with an unrun experiment.

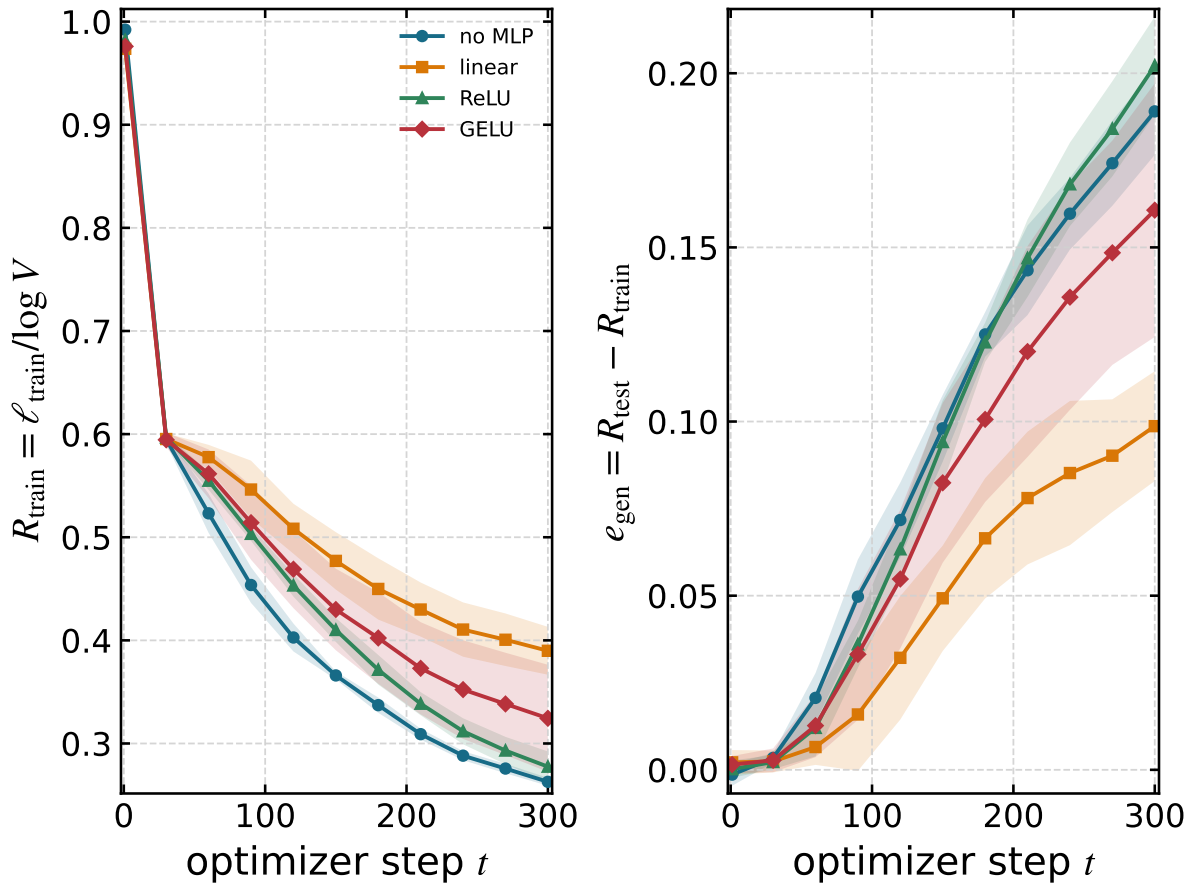


Figure 19: Training risk and generalization gap are separate $O(1)$ observables. Bands are seed-level 95% intervals.

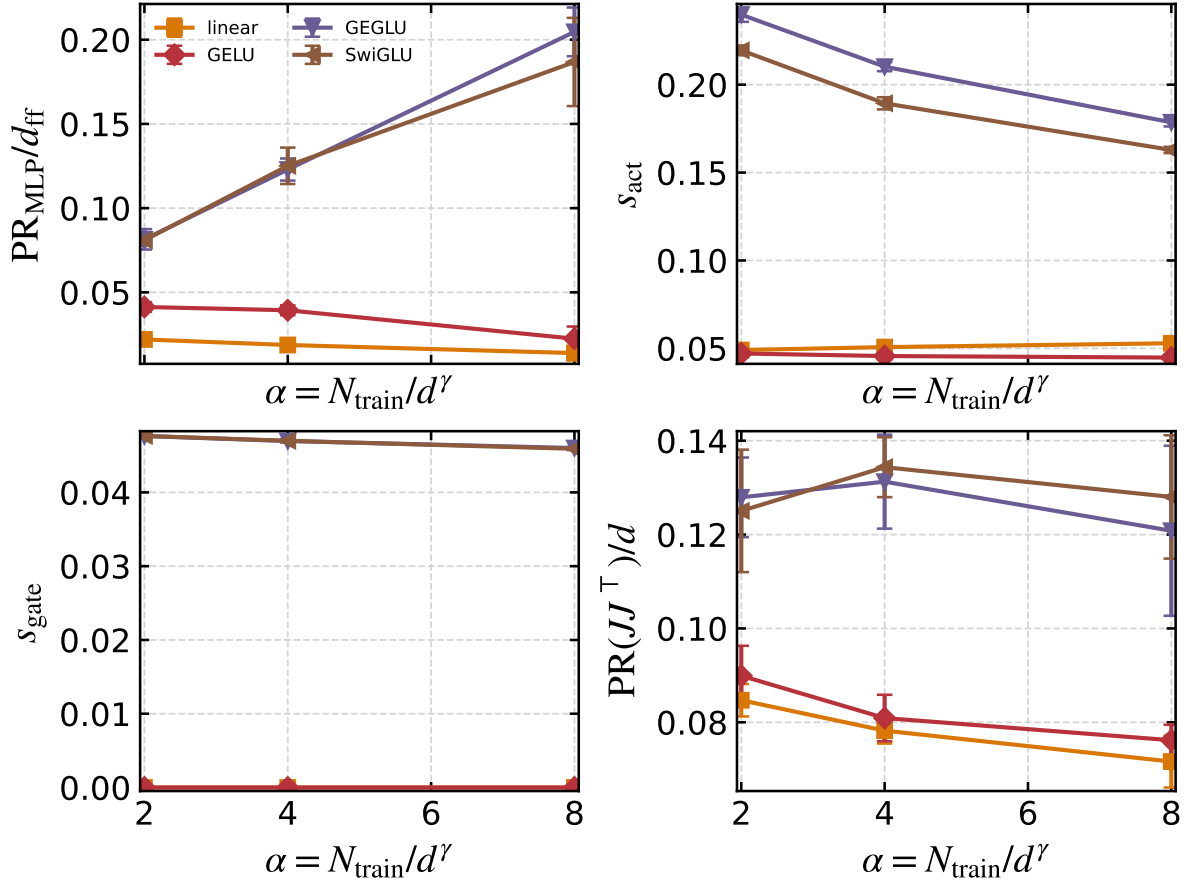


Figure 20: MLP mechanism confirmation. Spectral rank, activation occupancy, and gate statistics retain distinct names and interpretations.